

A search for transient unresolved optical features in public HARPS spectra

The single-exposure TOI-2458 case

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ABSTRACT

Context. Narrow, single-epoch features in high-resolution stellar spectra can arise from short-lived emission, night-sky lines, energetic-particle events, detector defects, or extraction artifacts; apparent significance alone is therefore insufficient for classification.

Aims. We searched a small public HARPS sample for variable unresolved optical features and performed a detector-level, all-epoch, and cross-instrument investigation of the strongest retained case, TOI-2458 (HD 34562; TIC 449197831).

Methods. We continuum-normalized, aligned, and differenced two HARPS epochs for each of ten targets from ESO programme 106.21TJ.001, then screened peaks at an absolute local residual threshold of 8 by width, morphology, spectral location, common-line proximity, and observer-frame recurrence. The strongest case was tested in extracted orders, the raw frame, three HARPS pipeline 3.6.0 reductions, all 88 public TOI-2458 HARPS products, and four earlier CHIRON and TRES spectra.

Results. The screen yielded 1534 threshold crossings; 13 passed automated checks and five remained unresolved one-exposure spikes after review. Two occurred together in one TOI-2458 exposure at archive barycentric air wavelengths 5761.26 and 6432.95 Å, with 19.80% and 10.94% excess. Both appear in extracted orders and survive controlled reductions, but lie in cosmic-mask-active detector neighborhoods. No credible recurrence occurs among the other 87 HARPS products: the only threshold-passing comparison event is narrower than a HARPS resolution element. No individual CHIRON or TRES comparison reaches $|z_{\text{emp}}| = 3$. For persistence at full event strength, expected CHIRON stacked z values are 5.61 and 4.97, versus observed values of 1.52 and 0.48.

Conclusions. Independent spectra disfavor features persistent at the recorded amplitudes. Simultaneous occurrence, cosmic-mask activity, absent credible recurrence, and a sub-resolution comparison event favor an exposure-specific detector or reduction origin, although no unique mechanism is established. Because the external spectra predate the event by about one year, they do not exclude a short natural episode. Classification requires expert inspection of the original two-dimensional frame or recurrence with an independent spectrograph.

Key words. instrumentation: spectrographs – methods: data analysis – techniques: spectroscopic – stars: individual: TOI-2458 – stars: activity – surveys

1. Introduction

High-resolution echelle spectra contain a mixture of stellar photons, atmospheric emission and absorption, calibration light, detector events, and products of extraction and wavelength calibration. Narrow positive excursions are especially difficult to interpret when they occur in only one exposure. A real unresolved emission line is broadened by the instrumental line-spread function, but a detector event can sometimes acquire a similar profile after spatial extraction. Conversely, a simple width cut can reject many obvious defects without identifying every remaining event.

HARPS is a fibre-fed, environmentally stabilized echelle spectrograph on the ESO 3.6-m telescope at La Silla. It covers approximately 3780–6910 Å at a resolving power near 115000 in the products used here (Mayor et al. 2003). Its public archive is therefore suitable for testing conservative searches for unresolved, time-variable optical features. This study is not a blind estimate of the incidence of such events. It is a deliberately small screening experiment followed by an intensive case study intended to show what can and cannot be inferred from a striking single-exposure residual.

The focal target is TOI-2458, an F-type star leaving the main sequence. Šubjak et al. (2025) reported 87 acquired HARPS spectra, a stellar effective temperature of 6005 ± 50 K, and a roughly 54-day activity cycle visible in the Ca II S-index and H α time series. The ESO archive query defined below returned 88 distinct processed spectra. We retained all 88 rather than imposing an undocumented rule to reconcile the one-product difference.

We first describe the small programme-level screen, then the complete HARPS and detector-level checks for TOI-2458, and finally an independent comparison with CHIRON and TRES spectra. Throughout, quoted local z values are descriptive noise-normalized measurements. They are not treated as globally calibrated false-alarm probabilities because the residuals are dominated by non-Gaussian stellar and instrumental structure.

2. Data

2.1. Initial HARPS sample

The screening sample contains 20 public one-dimensional HARPS products from ESO programme 106.21TJ.001: two selected epochs for each of ten targets. Table A.1 gives every product identifier and observation time. The products have nominal

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resolving power $R = 115000$ and span the optical HARPS range. This sample was selected for manageable manual review and is neither complete nor statistically representative of the HARPS archive.

2.2. Complete TOI-2458 HARPS sequence

After the two TOI-2458 features were retained, we queried the ESO Observation TAP service for public calibrated spectra satisfying all of the following: target name TOI-2458, instrument name beginning with HARPS, calibration level at least 2, spectral resolving power at least 100000, and data-product type “spectrum.” The query returned 88 unique product identifiers. Observation times, proposal identifiers, signal-to-noise metadata, source URLs, local filenames, byte counts, and SHA-256 checksums are provided in the supplementary manifest `data/toi2458_all_epochs/manifest.csv`.

The event product is `ADP.2022-02-20T01:06:53.624`, derived from raw exposure `HARPS.2022-02-19T00:15:10.071`. A second processed exposure, `ADP.2022-02-20T01:06:53.630`, began 1.69 h later and contains neither feature at comparable strength. It was used as the feature-free HARPS template in the cross-instrument analysis.

2.3. Raw and ancillary HARPS products

The ESO ancillary package for the event exposure supplies extracted `e2ds_A` orders and associated calibration products. The original raw detector frame and same-night order-position, flat-field, ThAr/ThAr, and ThAr/Fabry–Pérot calibration files were also retrieved. The raw exposure was reduced with ESO HARPS pipeline 3.6.0 in three otherwise matched configurations:

1. Laplacian cosmic-ray rejection disabled and extraction-stage k -sigma rejection disabled;
2. Laplacian rejection disabled and extraction rejection set to 3.5σ ; and
3. Laplacian rejection enabled with the documented 4/4/5 parameters and extraction rejection set to 3.5σ .

These controls test whether the two features are created or removed by the explicit rejection stages. They do not exhaust every possible detector or extraction failure mode. The Laplacian method is described by van Dokkum (2001).

2.4. CHIRON and TRES spectra

Four public spectra of TIC 449197831 were downloaded from ExoFOP-TESS: two CHIRON observations from the SMARTS 1.5-m telescope at CTIO and two TRES observations from the FLWO 1.5-m telescope. CHIRON was operated near $R = 80000$ and TRES near $R = 44000$ (Tokovinin et al. 2013; Szentgyorgyi & Fűrész 2007). FITS target identifiers and order coverage were validated before analysis. Table B.1 gives the exact file identifiers, observation times, resolving powers, quoted signal-to-noise ratios, and file checksums. All four observations cover both wavelengths of interest and predate the HARPS event by approximately one year.

3. Methods

3.1. Continuum normalization and epoch differencing

For each initial HARPS product, the continuum was estimated as the median positive flux in successive $5 \text{ \AA}$ bins and linearly interpolated to the native wavelength samples. Flux was divided by this estimate. Two spectra of the same target were aligned by minimizing the median absolute deviation of their difference over every twentieth valid sample, allowing a relative velocity shift between -2 and $+2 \text{ km s}^{-1}$.

The aligned normalized spectra were subtracted. A 40-pixel Gaussian-smoothed version of the raw difference was removed to suppress broad residual structure. The local scale $\sigma(\lambda)$ was estimated as 1.4826 times the median absolute deviation in $20 \text{ \AA}$ bins, interpolated across wavelength, with a conservative floor equal to half the tenth percentile of the valid bin scales. The local residual statistic was

$$z(\lambda) = \frac{f_2(\lambda) - f_1(\lambda) - b(\lambda)}{\sigma(\lambda)}, \quad (1)$$

where $b(\lambda)$ is the broad residual term.

Candidate peaks required $|z| \geq 8$, prominence of at least 5.2, separation of at least three samples, and a fitted half-prominence width no larger than 14 samples. The brighter epoch also had to contain a positive local excess rather than merely a shallower absorption line.

3.2. Automated rejection and manual review

The instrumental full width at half maximum (FWHM) was approximated by λ/R . A candidate was labelled as likely detector- or particle-like if its measured width was below 0.75 instrumental FWHM, and as a broad stellar-profile change if wider than 1.8 FWHM. Wavelengths outside 4000–6800 $\text{\AA}$ were labelled low-sensitivity spectral edges. We also rejected bipolar residuals characteristic of small alignment or profile changes, ordinary absorption-line variability, features within $0.20 \text{ \AA}$ of the Na I D lines, $H\alpha$, or the strong [O I] night-sky lines at 5577, 6300, and 6364 $\text{\AA}$, and observer-frame features recurring within $0.05 \text{ \AA}$ in unrelated targets.

Thirteen features survived these rules. Their local profiles and additional same-target spectra were then inspected manually. Candidate observer-frame wavelengths were also compared with the 2810-line UVES night-sky atlas of Hanuschik (2003). Absence from that atlas is evidence against a catalogued sky line, not proof that atmospheric contamination is impossible.

3.3. Injection and recovery

To characterize the limited sensitivity of the initial screen, unresolved Gaussian features with FWHM $\lambda/115000$ were injected at 12 randomly selected valid continuum locations in each of the ten pairwise residuals. Amplitudes were 0.02, 0.05, and 0.10 of the normalized continuum, giving 120 trials per amplitude. Recovery required a local maximum of at least 8σ and at least two samples of 5σ or greater near the injected position. This exercise measures only this implementation on these spectra; it is not a general HARPS completeness estimate.

3.4. Same-target recurrence tests

For each manually reviewed feature, the peak excess was measured within two instrumental FWHM of its selected barycentric wavelength. The local baseline used samples from three to eight FWHM on either side. A comparison exposure was counted as recurrent if its excess exceeded both 5% of the continuum and half the original event excess, and its peak lay within one instrumental FWHM of the event peak.

For TOI-2458, both wavelengths were subsequently fixed before the complete 88-product scan. Each other exposure was tested in both the stellar barycentric frame and the fixed observer frame. A recurrence additionally required local significance of at least 5σ . This fixed-location test avoids searching the comparison spectra for whichever peak happens to be strongest.

3.5. Detector-coordinate and pipeline tests

The two TOI-2458 features were located in the `e2ds_A` orders and mapped to zero-based detector coordinates. We checked whether each was already present before the final one-dimensional order merge, whether the simultaneous reference fibre had a comb maximum at the same dispersion position, and how the event intersected the pipeline cosmic-ray map (`CRH_MAP`). The count of flagged aperture pixels within ± 15 dispersion pixels was compared with identically sized windows elsewhere in the same order.

The 5761 Å feature's raw cross-dispersion profile was compared with the normal stellar trace using the Pearson correlation coefficient. A high correlation argues against one isolated detector pixel but cannot distinguish photons following the object trace from a particle track or extraction disturbance that overlaps it.

3.6. Wavelength conventions

The archive table describes its wavelength axis with `TUCD1=em.wl;obs.atmos` and `SPECSYS=BARYCENT`, indicating barycentric air wavelengths. The current pipeline array is explicitly named `WAVEDATA_VAC_BARY`. We converted the two archive air wavelengths to standard-air vacuum wavelengths using the refractive-index relation of Ciddor (1996). Comparison of labels was always made at the same detector pixel; it was not interpreted as a velocity shift of the source.

3.7. Independent-spectrum comparison

The relevant CHIRON or TRES order was continuum-normalized and aligned to the feature-free HARPS exposure by high-pass cross-correlation. The candidate region within ± 1 Å was excluded from alignment. Trial velocities ranged from -120 to $+120$ km s $^{-1}$ in 0.05 km s $^{-1}$ steps.

The event and control HARPS profiles were convolved from $R = 115000$ to the resolving power of the external spectrum, using Gaussian widths added in quadrature. The normalized external order was then differenced against the degraded control. A quadratic local trend was fitted in the interval 0.5 – 8 Å from the candidate, with five iterations of 3σ clipping. At the candidate position, we fitted the amplitude of a Gaussian whose FWHM equalled one external-instrument resolution element.

An empirical noise distribution was constructed by applying the same matched filter every 0.2 Å at placebo positions 1.0 – 7.5 Å on both sides. If a_0 is the candidate amplitude, $\tilde{a}_p$ the me-

Table 1. Automated classification of the 1534 initial peaks.

Classification	Number
Low-sensitivity spectral edge	782
Stellar absorption variability	472
Too narrow relative to HARPS resolution	226
Alignment or line-profile residual	18
Broad stellar line-profile change	14
Common-line proximity	6
Observer-frame recurrence, unrelated targets	3
Retained for manual review	13
Total	1534

dian placebo amplitude, and $s_p = 1.4826 \text{ MAD}(a_p)$, the reported statistic is

$$z_{\text{emp}} = \frac{a_0 - \tilde{a}_p}{s_p}. \quad (2)$$

The full-strength expectation was obtained by applying the same filter to the resolution-matched difference between the HARPS event and control spectra. An absolute empirical value of 3 was set in advance as the threshold for further manual review. Two observations from the same instrument were also combined with inverse-variance weights $1/s_p^2$. These stacks are transparent local diagnostics, not calibrated global false-alarm probabilities.

4. Results

4.1. Screening results

The initial pairwise search found 1534 peaks above the local threshold. Table 1 shows their automated classifications. Only 13 (0.85%) passed all automated checks. Manual inspection classified eight of those as ordinary stellar-profile or absorption-line changes and retained five as unresolved one-exposure spikes.

The five manually retained spikes are listed in Table 2. None coincides with a line in the checked UVES night-sky tables. The two TOI-2458 features occur in the same exposure; the two TOI-1054 features likewise occur together in one exposure. This clustering is important because an exposure-specific disturbance can produce multiple features simultaneously.

The injection experiment recovered 0/120 trials at 2%, 0/120 at 5%, and 59/120 (49.2%) at 10% of the continuum. Completeness varied substantially among targets: at 10%, individual pair recoveries ranged from 25% to 91.7%. The screen is therefore incomplete even for lines similar in amplitude to the weaker TOI-2458 event and supplies no useful constraint on a weak-feature population.

4.2. TOI-2458 event measurements

The two initial epoch-difference peaks had $|z| = 14.30$ and 8.80 . Re-measuring the event exposure alone against immediate sidebands gave local significances of 6.28 and 9.99. The values differ because the first statistic is based on an aligned, broad-trend-subtracted difference between epochs, whereas the second uses only the local scatter of one exposure.

Both features are visible in their original extracted orders, so the final order merge did not create them. At 5761 Å, the raw cross-dispersion profile correlates at $r = 0.98255$ with the

Table 2. Unresolved one-exposure spikes retained after manual review.

Target	Air wavelength (Å)	Excess (%)	Comparison spectra	Recurrence result
TOI-2458	5761.26	19.80	87	None significant
TOI-2458	6432.95	10.94	87	One event, but sub-resolution
TOI-1054	6109.82	11.35	2	None
TOI-1054	6116.58	14.03	2	None
TOI-282	6614.78	21.13	3	None

Notes. Wavelengths use the archive barycentric-air convention. Excess is measured above the local spectrum.

Table 3. TOI-2458 event and detector diagnostics.

Quantity	5761.26 Å feature	6432.95 Å feature
Event-exposure normalized peak	1.1942	1.1126
Local baseline	0.9962	1.0031
Excess	19.80%	10.94%
Single-exposure local significance	6.28 σ	9.99 σ
Initial epoch-difference $ z $	14.30	8.80
Extracted-order row, zero based	53	64
Detector x coordinate, zero based	1245	1506
Vacuum label at same pixel, pipeline 3.6.0	5762.863443 Å	6434.724733 Å
Re-reduced excess, rejection disabled	18.98%	10.56%
Re-reduced excess, k-sigma only	19.41%	10.56%
Re-reduced excess, k-sigma plus Laplacian	19.41%	12.26%
CRH_MAP flags at exact coordinate	0	1
CRH_MAP flags within ± 15 pixels	2	19
Same-order percentile of local flag count	99.803	97.196
Credible repeat in 87 other products	No	No

ordinary spatial profile of the stellar trace. Neither detector position is coincident with a reference-fibre comb maximum. These observations weaken the simplest one-hot-pixel explanation, but they do not eliminate a particle track, charge-spreading event, scattered light, local detector behavior, or an extraction failure.

The event neighborhoods contain unusual cosmic-mask activity. Near 5761 Å, two flagged pixels occur within the ± 15 -pixel window, placing the count at the 99.803rd percentile of windows in that order. Near 6433 Å, 19 flagged pixels occur in the window, including one at the event dispersion coordinate; most lie on one detector row. The count is at the 97.196th percentile. These are warning signs rather than categorical classifications because the extracted features remain after all three controlled reductions.

4.3. Complete HARPS recurrence test

At 5761.26 Å, none of the other 87 products passed the fixed recurrence threshold in either the stellar or observer frame. The event exposure ranks first by excess.

At 6432.95 Å, product ADP.2022-03-09T01:01:43.431 passed in both frames and has greater local excess than the original event. Its measured half-maximum width is only 0.020 Å, compared with an instrumental FWHM of approximately 0.05594 Å at that wavelength. A real monochromatic input is convolved with the line-spread function and cannot be substantially narrower than one resolution element. We therefore do not treat this second product as credible astrophysical recurrence; instead, it demonstrates that artifact-like spikes can pass an amplitude and local-significance cut at the same nominal wavelength.

4.4. Air and vacuum wavelength reconciliation

The apparent same-pixel offsets between the archive products and the new reductions are approximately 83 km s⁻¹. They are explained by wavelength convention. The Ciddor relation maps 5761.260000 Å in air to 5762.857943 Å in vacuum, compared with the pipeline same-pixel label 5762.863443 Å, a residual of +0.286 km s⁻¹. It maps 6432.950000 Å to 6434.728081 Å, compared with 6434.724733 Å from the pipeline, a residual of -0.156 km s⁻¹. Thus there is no unexplained 83 km s⁻¹ calibration displacement or source motion.

4.5. CHIRON and TRES comparison

No individual external measurement reaches $|z_{\text{emp}}| = 3$ (Table 4 and Fig. 3). Each CHIRON spectrum had sufficient local sensitivity to recover a persistent feature at the full HARPS-event strength at 3.3–4.2 empirical standard deviations. The observed CHIRON values are only 0.18–1.12. TRES expected values are 0.74–2.24, so its individual non-detections are not decisive.

The two CHIRON 5761 Å residuals are positive and each recovers approximately 27% of the full HARPS-event amplitude. Their individual significances are only 1.02 and 1.12, and their inverse-variance combination is 1.52; this is not a detection. Table 5 gives all four stacked diagnostics. A persistent full-strength model differs from the observed CHIRON stacks by -4.09 and -4.49 empirical standard deviations at 5761 and 6433 Å, respectively. Because the noise scale is local and the number of external epochs is only two per instrument, these values should be read as strong diagnostics against persistence at the original amplitude, not as formal population-level exclusions.

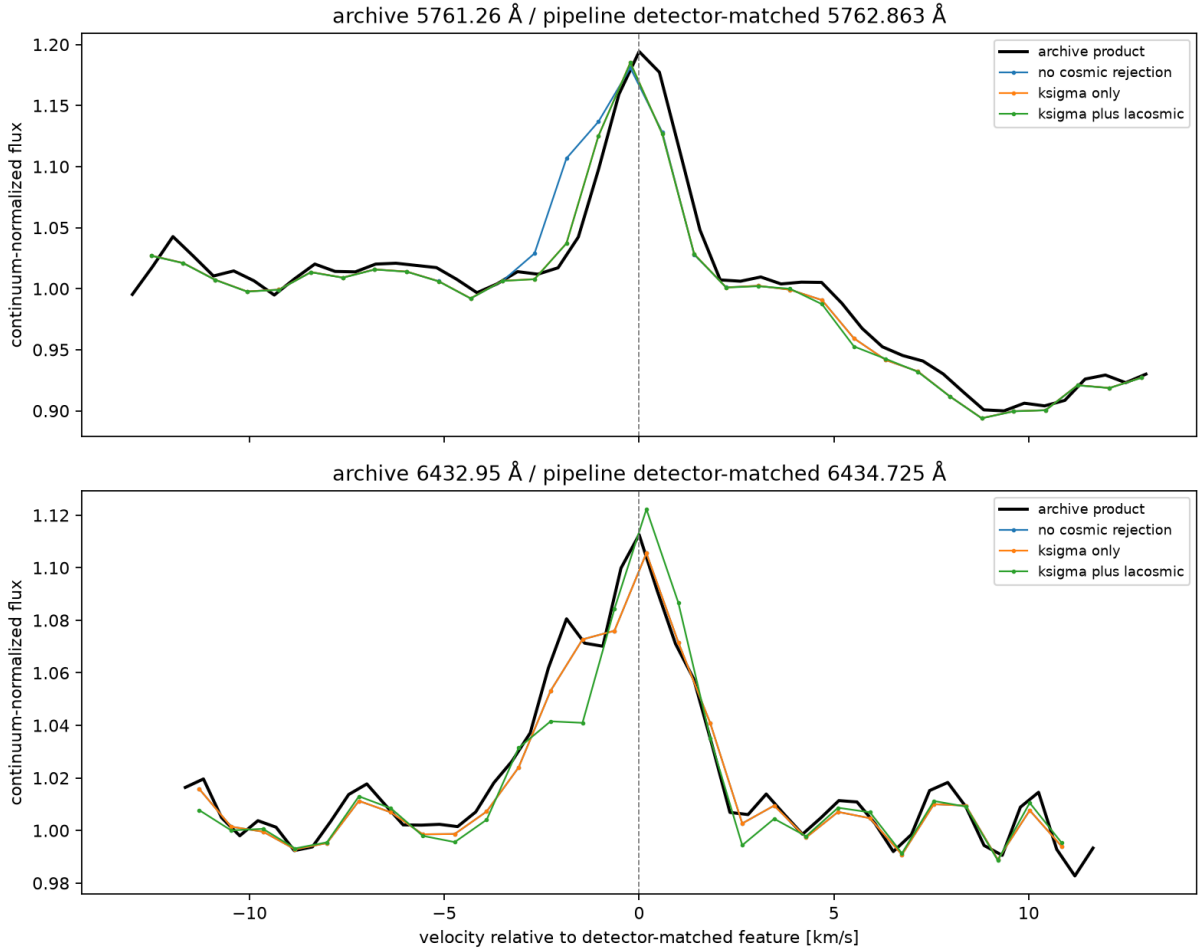


Fig. 1. Profiles of the two TOI-2458 events in the archive product and three controlled ESO HARPS pipeline 3.6.0 reductions. The black curve is the archive product; colored curves are detector-matched reductions under the three cosmic-rejection configurations. The different wavelength labels reflect the air/vacuum convention discussed in Sect. 4.4; curves are matched by detector coordinate.

Table 4. Independent-spectrum matched-filter measurements.

Instrument	FITS date (UTC)	λ_{air} (Å)	Observed z_{emp}	Full-strength expected z	Recovered fraction
CHIRON	2021-02-19 01:37	5761.26	1.02	3.76	0.272
CHIRON	2021-02-19 01:37	6432.95	0.18	3.69	0.050
CHIRON	2021-03-03 23:58	5761.26	1.12	4.16	0.270
CHIRON	2021-03-03 23:58	6432.95	0.51	3.32	0.153
TRES	2021-02-09 04:36	5761.26	-0.39	0.74	-0.529
TRES	2021-02-09 04:36	6432.95	0.28	1.62	0.173
TRES	2021-02-26 03:52	5761.26	-0.97	1.18	-0.819
TRES	2021-02-26 03:52	6432.95	1.71	2.24	0.763

5. Discussion

5.1. What is established

The excess counts are real features of the recorded event exposure. They are present in the extracted echelle orders, not introduced during the final one-dimensional merge, and they survive three reductions spanning disabled and enabled explicit cosmic-ray rejection. The archive-to-pipeline wavelength-label difference is fully accounted for by the air/vacuum convention. These conclusions are directly reproducible at fixed product and detector identifiers.

The data also establish that neither feature behaves as stable emission at its 2022 strength. The 5761 Å event does not

recur significantly in 87 other public HARPS products. The only threshold-passing 6433 Å comparison event is narrower than HARPS can resolve. Earlier CHIRON observations had adequate sensitivity to show both features if continuously present at full strength, but did not.

5.2. Most likely interpretation

Several independent observations favor an exposure-specific detector or reduction event: two features appear together; their detector neighborhoods contain unusually many cosmic-mask flags; one neighborhood includes a flag at the exact dispersion coordinate; the sole apparent recurrence is sub-resolution; and

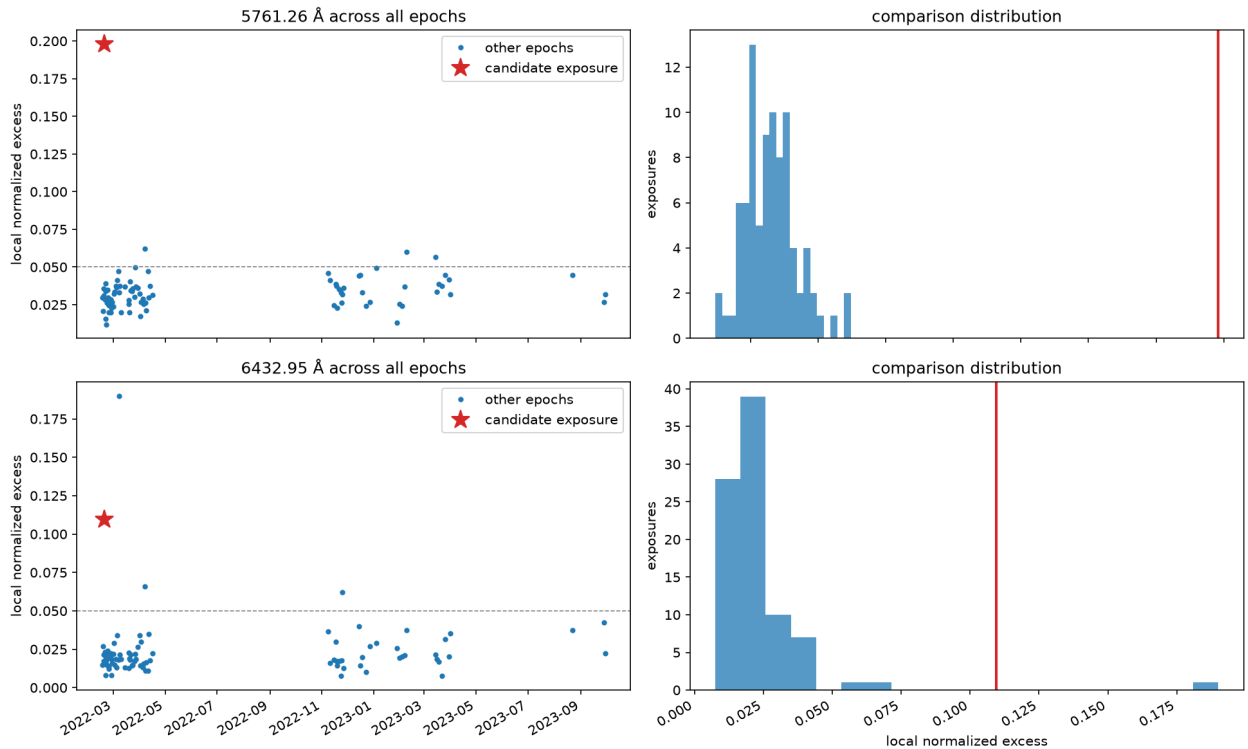


Fig. 2. Local excess at the two fixed wavelengths in all 88 public TOI-2458 HARPS products. Red symbols mark the original event exposure. The isolated large 6432.95 Å comparison value is the sub-resolution event described in the text.

Table 5. Inverse-variance cross-instrument stacks.

Instrument	λ_{air} (Å)	Observed stacked z	Full-strength z	Difference	Recovered fraction
CHIRON	5761.26	1.52	5.61	−4.09	0.271
CHIRON	6432.95	0.48	4.97	−4.49	0.096
TRES	5761.26	−1.03	1.40	−2.43	−0.737
TRES	6432.95	1.55	2.77	−1.22	0.560

no full-strength counterpart appears in sensitive CHIRON data. The 6433 Å mask morphology, with many flags concentrated along one detector row, is particularly suggestive of a particle or detector disturbance.

The evidence is not sufficient to select a unique mechanism. The 5761 Å cross-dispersion profile closely follows the object trace, both features survive pipeline rejection, and neither is a simple final-merge artifact. A short-lived stellar emission episode is therefore not logically excluded. TOI-2458 is known to be active on a roughly 54-day timescale, but the two wavelengths were not identified with established activity diagnostics, and the present data do not provide a physical line identification. “Unexplained transient” is consequently a more accurate observational description than a source classification.

5.3. Why the independent non-detections are temporally limited

All four external spectra were recorded in February or March 2021, whereas the HARPS event occurred on 19 February 2022. Their non-detections answer a persistence question: the star did not show stable features of the same full amplitude in those earlier observations. They cannot decide whether the 2022 exposure contains a short natural event or a one-exposure detector event, because both hypotheses predict absence one year earlier.

A decisive external confirmation would require contemporaneous exposures or a future recurrence observed with a different instrument.

5.4. Statistical limitations

The raw peak count shows why ideal Gaussian tail probabilities would be misleading. At an 8σ nominal local threshold, 1534 peaks were produced by spectral edges, absorption-line variability, alignment residuals, and detector-scale structure. The local z values are useful ranking statistics but do not encode the frequency of these nuisance processes.

Additional limitations are as follows:

1. The initial 20-spectrum sample is small and deliberately selected, so no event-rate inference is justified.
2. Comparing two epochs suppresses perfectly constant features and emphasizes any spectral or reduction change between them.
3. Continuum placement, alignment, and manual classifications can interact near deep or crowded stellar lines.
4. The injection experiment used only 120 trials per amplitude and does not span line widths, stellar types, or detector locations comprehensively.
5. The all-epoch test is complete only for public HARPS products satisfying the fixed archive query as of 15 August 2026.

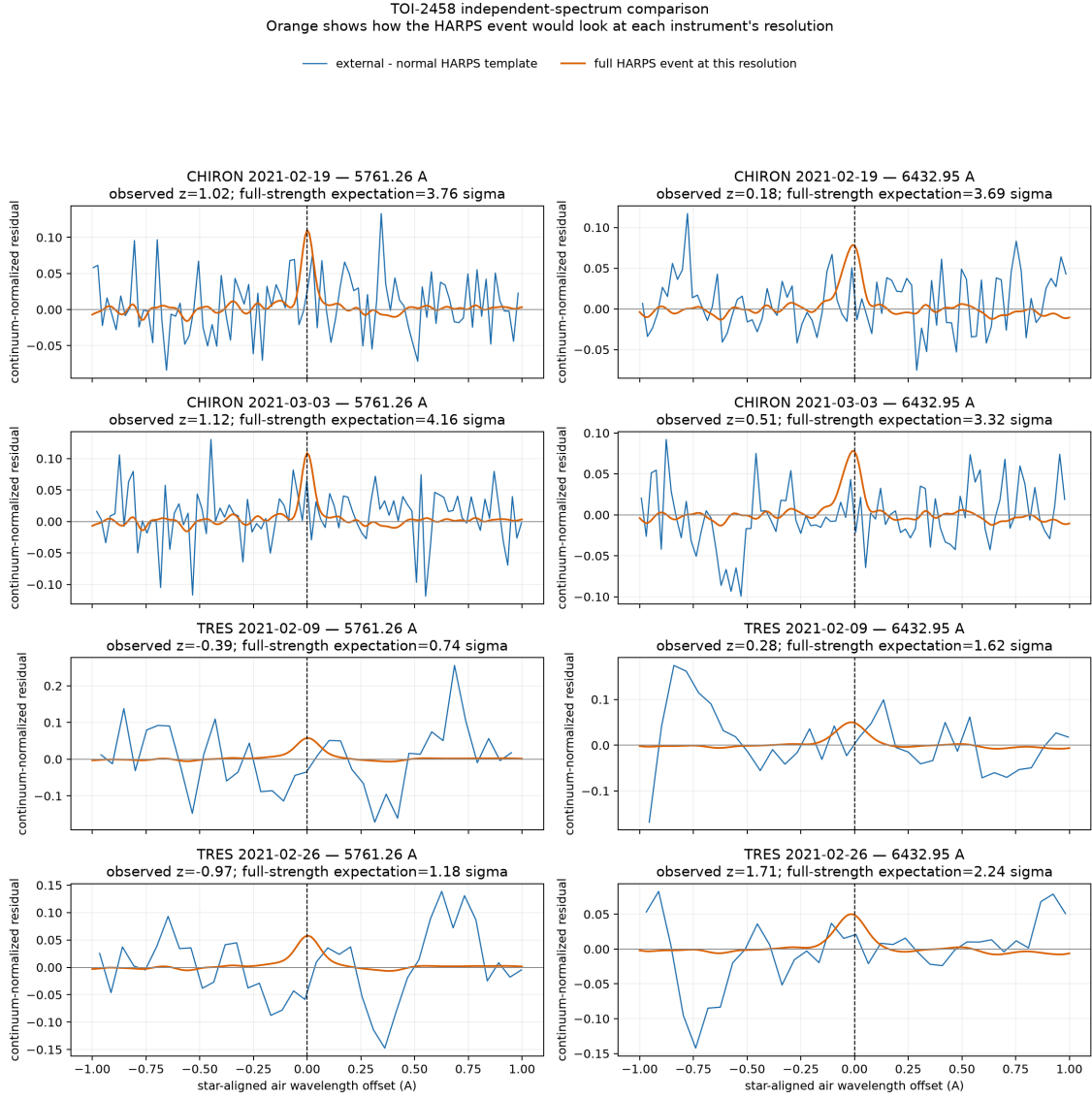


Fig. 3. Cross-instrument comparison at both event wavelengths. Orange curves show the full HARPS event after resolution matching; blue curves show the external spectrum minus the feature-free HARPS template. The dashed line marks the matched-filter center. The visible TRES fluctuations illustrate its limited sensitivity.

6. The external stack uses a local placebo scatter rather than a globally calibrated likelihood.
7. The raw-data controls test explicit rejection settings, not every stage of charge transport, calibration, tracing, scattered-light subtraction, and optimal extraction.

6. Conclusions

We conducted a conservative search for narrow variable features in 20 public HARPS spectra from ten targets and investigated the strongest retained case with substantially more data. The main conclusions are as follows:

1. Automated screening reduced 1534 local threshold crossings to 13 manual-review cases; five remained unresolved one-exposure spikes.
2. Two spikes occurred together in one TOI-2458 exposure at archive barycentric air wavelengths 5761.26 and 6432.95 Å, with local continuum excesses of 19.80% and 10.94%.

3. Both features exist in the extracted orders and survive controlled ESO pipeline 3.6.0 reductions, so they are not final-merge artifacts and are not removed by the tested cosmic-ray settings.
4. Cosmic-mask activity near both coordinates, simultaneous occurrence, and a sub-resolution apparent recurrence favor a detector or reduction origin without proving a specific mechanism.
5. The complete 88-product TOI-2458 sequence contains no credible recurrence at either wavelength.
6. Two CHIRON and two TRES spectra provide no independent match. CHIRON specifically disfavors features persistent at the full HARPS-event amplitudes; TRES is less sensitive.
7. The external observations predate the event by about one year and therefore cannot distinguish a short-lived natural emission episode from a one-exposure instrumental event.

The most informative next action is independent inspection of the original HARPS two-dimensional frame, order traces, ex-

traction weights, and cosmic-ray map at the coordinates in Table 3. If no instrumental mechanism is identified, future high-cadence observations should use an independent high-resolution spectrograph with consecutive exposures and simultaneous sky or calibration information.

Data availability and reproducibility

All astronomical source data are public. HARPS products are available from the ESO Science Archive by the product and raw-exposure identifiers given in this paper and its supplementary manifests. The CHIRON and TRES spectra are available from ExoFOP-TESS using the file identifiers in Table B.1. The supplementary package contains the complete manifests, per-feature measurements, recurrence tables, injection results, external matched-filter measurements, stacked diagnostics, figures, and SHA-256 checksums. Table C.1 maps each principal paper result to its machine-readable source.

The fixed selection rules and numerical procedures needed for an independent implementation are specified in Sect. 3 and summarized as an ordered, code-independent replication protocol in the accompanying `REPRODUCIBILITY.md`. No proprietary data or private catalogue access is required.

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Appendix A: Initial HARPS sample

Table A.1. Products used in the initial pairwise screen. Times are exposure start times in UTC.

Target	First product / time	Second product / time	Shift (km s ⁻¹)	Peaks	Review
HD 4915	ADP.2021-10-06T01:10:54.463 / 2021-10-05 03:55	ADP.2021-10-07T01:03:51.761 / 2021-10-06 04:26	-0.110	7	0
HD 39091	ADP.2020-11-18T01:18:40.358 / 2020-11-17 06:31	ADP.2020-12-03T01:10:58.554 / 2020-12-02 05:08	0.052	137	0
TOI-1054	ADP.2020-11-10T01:23:36.359 / 2020-11-09 00:52	ADP.2022-03-18T01:02:33.932 / 2022-03-17 09:40	-0.135	224	6
TOI-1203	ADP.2020-11-20T01:24:25.672 / 2020-11-19 08:12	ADP.2022-07-08T01:10:02.814 / 2022-07-06 22:54	0.108	143	0
TOI-1233	ADP.2020-12-23T01:06:55.617 / 2020-12-22 08:09	ADP.2021-05-26T01:01:31.042 / 2021-05-25 00:08	0.079	189	2
TOI-2194	ADP.2021-05-25T01:01:24.375 / 2021-05-24 06:06	ADP.2021-09-20T01:01:05.131 / 2021-09-19 00:37	-0.039	165	2
TOI-2458	ADP.2022-02-20T01:06:53.624 / 2022-02-19 00:15	ADP.2022-04-09T01:03:22.027 / 2022-04-07 23:24	0.116	136	2
TOI-261	ADP.2021-06-13T01:10:19.505 / 2021-06-12 09:23	ADP.2022-01-12T01:34:46.997 / 2022-01-08 01:34	0.136	187	0
TOI-282	ADP.2020-12-10T01:12:05.357 / 2020-12-09 03:04	ADP.2021-01-19T01:19:36.927 / 2021-01-18 02:18	0.194	152	1
TOI-4399	ADP.2021-09-17T01:03:12.717 / 2021-09-15 23:30	ADP.2022-07-08T01:10:02.854 / 2022-07-07 06:36	0.307	194	0

Notes. Shift is the fitted relative velocity. Review is the number sent to manual review.

Appendix B: Independent spectra

Table B.1. Public external spectra and integrity information.

Inst.	File ID	FITS date (UTC)	Filename	R	S/N	SHA-256
CHIRON	1041566	2021-02-19 01:37:01.4	TIC449197831S-ct20210219_1127.fits	80000	59.8	86a80943d93d9bcb1c47a4d10ac4096525fc5b106427d523569d387eb2d153f2d1b353bf8323cfe41e616d59ea0b532f0f05bce4f1ccba3a3427eed7dcbe867f83e27ebaa18ce416a0c019930c4368ca85a72d27d3735863bfd545f2e69dfa10670e959807576d1e54adfca03c349aa89ea8996341951e3fd5195b9779f902c
CHIRON	1041608	2021-03-03 23:58:22.3	TIC449197831S-ct20210304_1117.fits	80000	58.9	
TRES	197463	2021-02-09 04:36:41	TIC0449197831S-ab20210209.fits	44000	32.4	
TRES	200545	2021-02-26 03:52:33	TIC0449197831S-ab20210226.fits	44000	62.1	

Appendix C: Machine-readable result map and integrity record

Table C.1. Principal supplementary files. SHA-256 values describe the public-release version.

Content	Supplementary path	SHA-256
Initial 20-product input manifest	data/harps_106_21tj/manifest.csv	5ab71598ad25c527e8ad89c39799c8afd dfeb0444e424c64ea9b60b65c6c593ca38050c4bcc2b736211531331afb8c788a8ed3f211aa285dc485163c9179a4ef90ec77e3493df45d35e4b37f2bf6b2a50c4e1269f1c0ab699f1ceef0c27a7ab7e7788d501b7d7dc06bb6919d3c8ce2b3acd376fd44566ef1a5e056a97e33d670609ea1e36bad011306084dfcee236d2a83edac3d300d1ce5055991996e153cba44e458be0d05b919f16a06ce5308e2f38d333e892da4f264abf214a05837612de9d7a1fe32ce54edfa22d3cebb29d82b7d6a326529f4393e42cc8f7173a4d0bad048c750927071aa2476158d1aa213750d776fde83ebad76840338b56044bda8c04ae13b3d014dc08f6d347a07c62e9689f4706e03f25a2b62b3567ae4ff79a64fa1048996d0d6935342ae12f1817785ca58ce2853bcfa1d19032505dccc64b9c32e8d0fd4319439a65c8bcf677aa51675ec50cfcbdb622a71210ab9050dec b7cbc65a6aeb1febca28a3739f942dc21fff85cddb1ddd51d292a167cbc3ab43f5c715ebccdb769af23647beec9055c30b888b4e1d870142046fb5603b81
All 88 TOI-2458 products	data/toi2458_all_epochs/manifest.csv	
Four external products	data/toi2458_external_spectra/manifest.csv	
All 1534 screening peaks	results/harps_residual_search/candidates.csv	
Manual dispositions	results/harps_residual_search/manual_review_notes.csv	
Injection/recovery trials	results/harps_residual_search/injection_results.csv	
Initial recurrence checks	results/harps_residual_search/recurrence_results.csv	
Extracted-order checks	results/harps_residual_search/order_level_checks.csv	
Complete fixed-line scan	results/harps_residual_search/toi2458_all_epoch_line_scan.csv	
Detector and pipeline diagnostics	results/harps_residual_search/candidate_cosmic_check.json	
Individual CHIRON/TRES measurements	results/harps_residual_search/external_spectra_line_comparison.csv	
CHIRON/TRES stacks	results/harps_residual_search/external_spectra_stacked_summary.csv	
External-comparison figure	results/harps_residual_search/plots/external_spectra_comparison.png	